Coverage Debt and Adaptive Conformal Calibration after Structural Breaks

Cao Phi Phụng[1,*], Trần Quang Vinh[2]

[1]Ton Duc Thang University, Ho Chi Minh City, Vietnam

[2]Independent Researcher, Ho Chi Minh City, Vietnam

[*]Corresponding author: caophiphung@tdtu.edu.vn

ORCID: Cao Phi Phụng 0009-0001-5603-6569; Trần Quang Vinh 0009-0004-7105-4484

Abstract

Online conformal calibration is commonly evaluated by time-averaged coverage, which can obscure short periods of undercoverage after a structural break. We study this transient behavior using two break-conditional diagnostics: coverage debt (CD), the accumulated excess miscoverage after a break, and time-to-recovery of coverage (TTR-C). We give a conditional finite-horizon adaptation-lag bound for fixed-step Adaptive Conformal Inference (ACI) and propose BR-CP, a model-agnostic calibration wrapper that monitors the conformity-score stream with a Page-CUSUM, increases adaptation as evidence accumulates, resets stale calibration memory after an alarm, and uses a tolerance-inspired small-buffer order statistic while the post-break buffer refills. At break magnitude 3, averaged across seven families with 200 replications per family, BR-CP reduces mean CD from 2.97 for the best tested fixed-step ACI comparator to 2.00, compared with 1.54 for a benchmark that knows the break time. The complete simulation grid comprises three break magnitudes. Its empirical marginal coverage across the 21 break configurations ranges from 0.865 to 0.892 against a 0.900 target, with the largest shortfall under sustained trend changes. In controlled stress tests on five real-world benchmark datasets, using a frozen archived global forecaster whose training-corpus provenance was not independently reverified in this revision, BR-CP achieves CD 3.21 versus 3.82 for the best evaluated ACI and recovers coverage in 8 steps versus 27–55 for adaptive baselines, with no detected interval-score difference relative to tuned ACI in the reported comparison. In two descriptive, externally dated studies, the detector produces seven post-event alarms across five of eight exchange-rate channels during 260 daily observations after Lehman but no alarm in four influenza channels during 60 weeks after the H1N1 declaration; neither event causes a same-observation alarm. The results support break-conditional evaluation when transient post-break calibration matters.

Keywords: conformal prediction; structural breaks; change-point detection; uncertainty quantification; online calibration; time-series forecasting.

1 Introduction

Pretrained time-series forecasting models can be applied to a target series without target-specific parameter updates, as illustrated by Chronos, TimesFM, and Moirai [1–3]. When the forecaster is frozen, adaptation to distribution shift must occur outside the model. Online conformal methods such as Adaptive Conformal Inference (ACI) and Conformal PID provide a practical calibration layer because they update prediction intervals sequentially without retraining the forecaster [4, 5].

The usual summary metric is marginal coverage over a long evaluation period. That metric does not describe when miscoverage occurs. After a structural break, an adaptive conformal layer can under-cover for a short interval and later compensate with over-coverage, while its long-run coverage remains close to nominal. For applications in which decisions are concentrated around regime changes, this transient behavior is relevant even when the eventual time average is acceptable.

This paper evaluates post-break calibration directly. We use a break-conditional formulation of *coverage debt* (CD) to measure accumulated excess miscoverage and *time-to-recovery of coverage* (TTR-C) to measure

recovery time. An independently posted 2026 non-peer-reviewed manuscript by Peyton Kocher also uses the phrase "coverage debt" in a two-timescale conformal setting; we therefore make no priority claim for the terminology. We then study whether fixed-step tuning alone is sufficient and propose BR-CP, a break-responsive wrapper that monitors conformity scores rather than raw observations. The detector modifies only the calibration layer; the point forecaster remains frozen.

The closest recent work is CPTC, which explicitly addresses conformal forecasting with change points by combining online conformal prediction with a model that predicts latent state transitions [13]. BR-CP addresses a different operating constraint: it does not require a state-transition model and reacts only to changes visible in the conformity-score stream. The contribution is therefore not the first change-point-aware conformal method. It is a score-triggered calibration mechanism together with diagnostics that quantify transient post-break miscoverage.

Contributions.

1. **Break-conditional evaluation.** We define a break-conditional CD functional and TTR-C (Section 3) and show that they distinguish calibration methods whose marginal coverage is similar.
2. **Fixed-step analysis.** We state a conditional finite-horizon adaptation-lag bound for fixed-step ACI and show empirically that the recovery-efficiency trade-off is U-shaped over the evaluated step-size grid (Section 4).
3. **Break-responsive calibration.** BR-CP uses a Page-CUSUM on conformity scores, evidence-responsive step sizes, post-alarm memory reset, a tolerance-inspired small-buffer radius rule, and detector re-anchoring (Section 5). The method is model-agnostic and does not require gradients or forecaster retraining.
4. **Controlled and real-data evidence.** We evaluate seven simulated break families and conduct controlled stress tests on five public benchmark datasets with a frozen cross-domain forecaster. We also replay two externally dated event windows, finding heterogeneous and delayed alarms rather than an event-synchronous response; these descriptive results mark a boundary of applicability.

Scope. Sections 6–7 use controlled simulations with a frozen linear forecaster so that break times are known and comparisons are paired across 200 replications per cell. Section 8 uses a global forecaster trained on electricity, traffic, and weather series and then frozen when applied to ETT and exchange-rate benchmarks. Structural breaks in the main real-data arm are injected into observed series so that the break time is known; this arm is therefore a semi-synthetic stress test on real-world data rather than a natural-experiment study. The historical main experiment does not evaluate a published time-series foundation-model checkpoint; the separate post-hoc Chronos-Bolt Small analysis in Section 10 provides exploratory checkpoint-specific evidence only. No claim about TimesFM, Moirai, or training-corpus disjointness follows from the historical results.

2 Related work

Online conformal prediction for time series. Split conformal prediction relies on exchangeability [6, 7], which is generally violated by dependent time series. Sequential approaches include Sequential Predictive Conformal Inference (SPCI) [8], direct robustness analyses under non-exchangeability [9], and online-learning methods that adapt the calibration level. The latter include ACI [4], AgACI [10], strongly adaptive online learning [11], arbitrary-shift conformal inference [12], and Conformal PID [5]. BR-CP belongs to this online-calibration family but adds an explicit break-response mechanism.

Change points and conformal calibration. Sun and Yu's CPTC combines a state-transition model with online conformal prediction to improve adaptation when latent dynamics change [13]. Their method can exploit predicted state transitions before or during a shift. BR-CP instead uses no latent-state predictor; it monitors the realized conformity-score stream and reconfigures calibration memory when that stream changes. Vovk et al. use conformal test martingales to decide when a predictive model should be retrained [14]. Our detector is used for a different intervention: the forecaster remains fixed and only the calibration layer changes.

Change detection. Page's CUSUM is a classical sequential change-detection procedure [15], and Lorden's analysis relates false-alarm control to worst-case detection delay [16]. We use this theory as a reference for detector calibration rather than as a universal lower bound on BR-CP, because BR-CP begins adapting before a formal alarm when the CUSUM statistic rises.

3 Setting and break-conditional diagnostics

3.1 Online calibration of a frozen forecaster

Let $y_1, y_2, \dots$ be a real-valued series. A *frozen* forecaster f maps a context window to a one-step-ahead point forecast $\hat{y}_t = f(y_{t-L:t-1})$; its parameters never change. Write the conformity score $s_t = |y_t - \hat{y}_t|$. A calibration layer emits, before observing y_t, a radius q_t measurable with respect to $\mathcal{F}_{t-1} = \sigma(y_1, \dots, y_{t-1})$, giving the interval $C_t = [\hat{y}_t - q_t,\ \hat{y}_t + q_t]$ and the miscoverage indicator $\mathrm{err}_t = \mathbf{1}\{s_t > q_t\}$. The nominal level is $1 - \alpha$.

ACI maintains an internal level α_t and updates

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t), \qquad q_t = \widehat{Q}_{1-\alpha_t}(s_{t-W:t-1}), \tag{1}$$

where $\widehat{Q}$ is the corrected empirical quantile. The idealized unprojected recursion gives the telescoping identity $T^{-1}\sum_{t=1}^{T}(\mathrm{err}_t - \alpha) = (\alpha_1 - \alpha_{T+1})/(\gamma T)$, with the boundary convention specified by Gibbs and Candès [4]. The evaluated implementation instead projects α_t to $[0, 1]$ and uses a finite maximum radius if the target quantile exceeds the buffer resolution. Writing $r_t = \alpha_{t+1} - [\alpha_t + \gamma(\alpha - \mathrm{err}_t)]$, its exact identity is

$$\frac{1}{T}\sum_{t=1}^{T}(\mathrm{err}_t - \alpha) = \frac{\alpha_1 - \alpha_{T+1} + \sum_{t=1}^{T} r_t}{\gamma T}. \tag{2}$$

Therefore the unprojected ACI finite-time guarantee is *not* asserted for our clipped, finite-radius implementation. The same finite-radius convention is used for ACI experts and BR-CP; clipping/saturation effects remain a limitation of the reported comparisons.

3.2 The problem with marginal coverage

Suppose the score distribution jumps at an unknown time τ from F_0 to F_1, with F_1 stochastically larger. Immediately after τ the radius still reflects F_0, so miscoverage rises to $p_1 = 1 - F_1(Q^{F_0}_{1-\alpha}) \gg \alpha$. Equation (1) then drives α_t down until coverage is restored, and *overshoots*: the same integral action that repaid the deficit keeps pushing, producing a period of over-coverage that cancels the deficit in the time average. Marginal coverage is therefore not merely insensitive to the transient; it is insensitive *by design*.

3.3 Two diagnostics

Let $m_j = \mathbb{P}(y_{\tau+j} \notin C_{\tau+j})$ be the marginal miscoverage probability at break offset j. In the synthetic arm, the corresponding estimator averages the 0/1 miss indicator across the 200 paired independent seeds; in the real benchmark stress tests, it averages across the 322 injected-break windows (nested within 35 series).

Definition 1 (Coverage debt)**.** *For a horizon containing H offsets, indexed $j = 0, \dots, H-1$, and $\varepsilon \geq 0$, let*

$$\mathrm{CD}_{\varepsilon,H} = \sum_{j=0}^{H-1}(m_j - \alpha - \varepsilon)_+, \qquad (x)_+ = \max(x, 0). \tag{3}$$

CD is the sum of positive excess *marginal miscoverage probabilities*, not the expected positive-part sum of realized errors from a single path. Its plug-in estimator is nonlinear and is upward biased near the tolerance boundary; bootstrap intervals therefore describe uncertainty in the estimator, not a guarantee of conditional coverage.

Definition 2 (Time to recovery of coverage)**.** *Strict TTR-C is the smallest offset j at which $m_{j'}$, $j' \in \{j, \ldots, j+w-1\}$ all lie below $\alpha+\varepsilon$; the experimental persistence window is $w = 25$. The reported* marginal *TTR-C is instead the first j such that $P^{-1}\sum_{r=j}^{j+P-1} m_r \le \alpha+\varepsilon$, with seasonal period $P = 24$ for the displayed synthetic and real-data summaries. The latter is a single rolling-average threshold crossing, not a 25-step persistence condition. The CD horizon is $H_{\mathrm{CD}} = 200$ in both arms, but the* recovery *search horizons differ in the archived code: 500 post-break offsets in the synthetic arm and 200 in the real-data stress tests. Failure to recover within the relevant search horizon is right-censored (synthetic: > 500; real stress test: > 200), not a numeric event time; the archives may encode censoring using* `inf` *or a plotting sentinel such as 201.*

The tolerance used in the published simulation tables is $\varepsilon = 2\sqrt{0.1(0.9)/200} = 0.04243$; for the main real-data variance stress test, using 322 pooled windows gives $\varepsilon = 2\sqrt{0.1(0.9)/322} = 0.03344$. These thresholds reduce positive-part estimation noise but do not eliminate it. Applying the same ε to every method does *not* preserve ranking in general. We additionally report $\varepsilon = 0$ and a grid of other values below (Section 9). At $\varepsilon = 0.04243$, recovery means miscoverage at most 0.14243, not a guarantee of nominal 0.10 miscoverage.

The difference between strict and marginal TTR-C matters in the seasonal amplitude experiment. The evaluated unconditioned-radius methods can have periodic residual miscoverage despite acceptable period-averaged coverage; a suitable season-dependent score or scale normalization might address this, but was not tested here.

4 Limits of fixed-step tuning

Proposition 1 (Conditional adaptation-lag lower bound)**.** *Consider a fixed target internal level $\alpha^\star$ over the post-break horizon and a projected ACI recursion whose per-step change is bounded by $\gamma > 0$. Suppose its initial distance is a nonrandom $\Delta = |\alpha_\tau - \alpha^\star| > 0$. Assume, in addition, that the* marginal *post-break miss probabilities satisfy, for each $j = 0, \ldots, H-1$,*

$$m_j - \alpha \;\ge\; c(\Delta - (j+1)\gamma)_+ \quad \textit{for some constant } c > 0.$$

Then with $r = \min\{H, \max(0, \lceil \Delta/\gamma \rceil - 1)\}$,

$$\mathrm{CD}_{0,H} \;\ge\; c\left[r\Delta - \frac{\gamma r(r+1)}{2}\right]. \tag{4}$$

If $H \ge \lceil \Delta/\gamma \rceil - 1$ and $\gamma \downarrow 0$ with fixed Δ and c, the right-hand side equals $c\Delta^2/(2\gamma) + O(c\Delta)$.

Proof. Projection onto $[0,1]$ cannot enlarge the absolute change produced by one projected ACI update, so after $j+1$ steps its distance from the fixed target is at least $(\Delta - (j+1)\gamma)_+$. This reachability observation alone gives no bound on miscoverage. The explicitly assumed marginal miscoverage inequality provides the needed connection to the CD functional. Applying the positive part and summing through H offsets gives $c\sum_{i=1}^{r}(\Delta - i\gamma)$, which equals $c[r\Delta - \gamma r(r+1)/2]$. When $0 < \Delta \le \gamma$ the sum is zero. For $H \ge \lceil \Delta/\gamma \rceil - 1$ and fixed Δ as $\gamma \downarrow 0$, the arithmetic progression yields the stated asymptotic expansion. □

This is a *conditional explanation*, not a data-free lower bound for our rolling-buffer implementation: changing buffer contents can restore coverage before the internal level reaches a fixed $\alpha^\star$; the required miscoverage inequality and fixed target are not established there. No bound that grows without limit as $\gamma \to 0$ can apply to CD with a fixed H, since $\mathrm{CD}_{0,H} \le H(1-\alpha)$. In the evaluated grid, debt rises again at large γ; that observation is empirical and distribution-dependent rather than a convexity theorem.

Classical quickest-detection theory provides a separate reference point. For procedures that keep their calibration state unchanged until a detector fires, Lorden-type bounds imply that tighter false-alarm control generally increases the delay before a change can be identified [16]. If post-break miscoverage remains near p_1 during that delay, the corresponding pre-alarm contribution to coverage debt scales with $(p_1 - \alpha)$ times the delay. This is only a benchmark for passive-before-alarm schemes. It is not a lower bound for BR-CP, because the graded step size begins adapting before the alarm.

5 Break-responsive conformal prediction

5.1 Monitor the model, not the world

A calibration layer should react to changes that damage *the model's competence* and ignore changes that do not. Changes in the observation distribution and changes in forecast-error behavior need not coincide. Many modern forecasting pipelines normalize or scale context windows. Instance normalization, which is used in our controlled experiments, absorbs level and scale shifts by construction. In our simulation a 3σ level shift moves the observation distribution by 1.73 standard deviations while changing mean absolute error by 4%: a large, obvious event in the data that requires no recalibration whatsoever. Conversely, a drop in the AR coefficient with the innovation scale rescaled to preserve $\mathrm{Var}(y)$ exactly leaves the observation distribution untouched while inflating one-step errors by 119%.

The conformity-score stream is a direct monitoring target because it records the discrepancy that determines whether the current interval radius is adequate. We therefore run a two-sided Page-CUSUM on standardized, winsorized scores against a frozen anchor window $\mathcal{A}$:

$$z_t = \mathrm{clip}\left(\frac{s_t - \hat{\mu}_{\mathcal{A}}}{\hat{\sigma}_{\mathcal{A}}},\, -W_c,\, W_c\right), \tag{5}$$

$$S_t^+ = \left(S_{t-1}^+ + z_t - k\right)_+, \quad S_t^- = \left(S_{t-1}^- - z_t - k\right)_+, \tag{6}$$

with an alarm at $\inf\{t : \max(S_t^+, S_t^-) \geq h\}$. Winsorizing at $W_c = 4$ bounds the influence of a single heavy-tailed score; the clipping may also attenuate very large shifts. We also evaluate a rank-transformed increment $z_t = u_t - 1/2$, where u_t is a finite-anchor rank of s_t. The observed longer delays cannot be attributed to the numerical cap $|z_t| \leq 1/2$ alone: simultaneous rescaling of increments, slack k, and threshold h preserves alarm times. The standardized signal retains information about the magnitude of a score change that ranks discard, but the evaluated detectors also use different calibrated operating points.

5.2 Four break-response mechanisms

Between alarms BR-CP is ACI with a small step γ_{slow}. The detector then supplies four things.

(R1) Evidence-responsive step size. Before an alarm, the step interpolates with the normalized detector statistic,

$$\gamma_t = \gamma_{\mathrm{slow}} + (\gamma_{\mathrm{fast}} - \gamma_{\mathrm{slow}}) \min\left(1, \max(S_t^+, S_t^-)/h\right), \tag{7}$$

so adaptation accelerates as evidence accumulates. In the evaluated implementation, γ_t is also held at γ_{fast} for the first 100 steps after an alarm. The ablation in Section 7 shows that this post-alarm fast phase is not necessary for the observed gain; the memory reset is the dominant component. Main results nevertheless report the implementation that was evaluated, and the no-boost variant is shown separately.

(R2) Memory reset to Page's estimate. On an alarm the calibration buffer is truncated to the last $\max(n_{\min}, \hat{\delta})$ scores, where $\hat{\delta}$ is the run length since the CUSUM was last at zero, which is Page's classical change-point estimate. The stale pre-break scores that were pinning the quantile down are discarded outright.

(R3) Tolerance-inspired small-buffer rule. For an *i.i.d., continuous, fixed-size* sample from a common score distribution, $F(s_{(k)}) \sim \mathrm{Beta}(k, n-k+1)$ and a one-sided $(1-\alpha_t)$-content order statistic with confidence $1-\delta$ can be selected from $\mathrm{BetaCDF}(1-\alpha_t; k, n-k+1) \leq \delta$. Our implementation uses $\delta = 0.10$ and $n_{\min} = 15$. An alarm chooses the retained buffer adaptively and the real score stream is dependent, so *this nominal confidence does not carry over* to BR-CP. R3 is a *tolerance-inspired heuristic* for temporarily increasing the radius, not a finite-sample distribution-free coverage certificate for the selected post-alarm buffer. If no order statistic meets the nominal criterion, the code uses the observed maximum. Even under the ideal fixed-sample i.i.d. model, its confidence is only $1-(1-\alpha_t)^n$: at $n = 15$ and $\alpha_t = 0.1$ this is approximately 0.794, less than 0.90. Moreover, because α_t changes online, R3 cannot be interpreted as a fixed-target tolerance guarantee.

(R4) Re-anchoring. After an alarm the detector is disarmed while the next 60 scores form a new anchor. It does not detect changes within that period. This blind window is an observed limitation under closely spaced breaks.

Reset variant. The standard reported BR-CP resets the internal level to α on alarm. The evaluated *BR-CP-safe* variant instead uses $\alpha_{t+1} = \min(\alpha_{t+1}^{\mathrm{pre-reset}}, \alpha)$; it therefore does not increase the internal miss

Table 1: Implemented main-arm settings; main BR-CP results retain the 100-step post-alarm fast phase. Variants are labeled separately.

Parameter	Evaluated value
Nominal miss rate α	0.10
Calibration window W	300
$\gamma_{\text{slow}}, \gamma_{\text{fast}}$	0.01, 0.10
Post-alarm boost duration	100 steps (main method)
Minimum retained buffer $n_{\min}$	15 scores
Tolerance rule δ	0.10 (nominal; heuristic)
Detector anchor / rearm	250 / 60 scores
Synthetic detector (k, h)	(0.65,6.0)
Real-data detector (k, h)	(0.8,8.0)
Evaluation CD horizon	200 offsets
Strict TTR persistence / marginal period	25 / 24

level at the reset event. This does *not* prove that the emitted next interval is monotonically wider, because the score buffer also changes.

5.3 Coverage accounting

The unconditional long-run guarantee of fixed-step ACI does not transfer unchanged to BR-CP. The implementation combines clipped calibration levels, time-varying step sizes, explicit resets, and detector re-anchoring, so the usual telescoping argument for the unmodified ACI recursion is not sufficient to establish an unconditional guarantee. We therefore do not claim a new long-run coverage theorem for BR-CP.

Instead, long-run coverage is reported as an empirical diagnostic alongside break-conditional metrics. The aggregated mean over the 21 simulated break configurations is 0.8866, but this average conceals important heterogeneity: BR-CP's per-configuration coverage ranges from 0.8647 (trend, magnitude 4) to 0.8920 (level, magnitude 4), against a nominal 0.900. The separate no-break control has coverage 0.8922. In the trend family, coverage falls from 0.8792 at magnitude 2 to 0.8647 at magnitude 4. Thus the deficit is not uniformly around one percentage point: sustained trends produce markedly larger shortfalls. These quantities were checked directly against the archived `table_main.csv`, and are part of the method's empirical trade-off. This limitation is revisited in Section 12.

5.4 Detector calibration

The alarm threshold cannot be set from asymptotics because conformity scores are neither uniform nor independent. We calibrate (k, h) empirically on a grid, estimating the average run length to false alarm ARL_0 on a no-break DGP and the expected detection delay on three break families, and select the smallest expected delay subject to $\text{ARL}_0 \geq 1500$. The chosen operating point, $k = 0.65$, $h = 6.0$, yields $\text{ARL}_0 = 1882$ and expected delays of 17.9, 12.8 and 11.2 steps on level, variance and predictability breaks respectively. Repeating the same procedure for the rank-based statistic selects $k = 0.05$, $h = 5.0$ at a comparable $\text{ARL}_0 = 1600$, but delays of 60.8, 31.7 and 25.8 steps, two to three times slower. The standardized increment retains the magnitude of large residual deviations whereas the rank transformation discards that magnitude information. The raw numerical cap $|z_t| \leq 1/2$ is *not* an invariant explanation: rescaling the rank increments, slack and threshold together leaves the alarm times unchanged. The observed delay comparison is specific to the separately tuned operating points. This calibration uses only no-break and synthetic-break data and is never re-tuned per configuration.

One caveat applies to reading such a grid. Expected detection delay is only interpretable where ARL_0 is high: at a loose threshold a spurious alarm arriving shortly after τ is indistinguishable from fast detection and flatters the estimate. This is why the false-alarm constraint is imposed before, not after, selecting on delay.

Algorithm 1 One causal BR-CP update (main evaluated variant)

state: buffer B_{t-1}, level α_t, CUSUM S^+_{t-1}, S^-_{t-1} and run lengths r^+_{t-1}, r^-_{t-1}; armed flag, time since alarm, post-alarm fast-phase counter
Before observing y_t**:** issue $q_t = \widehat{Q}_{1-\alpha_t}(B_{t-1})$ or heuristic R3 radius when the post-reset buffer is refilling
Observe y_t and compute $s_t = |y_t - \hat{y}_t|$ and $e_t = \mathbf{1}\{s_t > q_t\}$
Set $S_{t-1} = \max(S^+_{t-1}, S^-_{t-1})$ and choose $\gamma_t = \gamma_{\text{fast}}$ if inside the 100-step boost; else choose the R1 interpolation using S_{t-1}/h when armed; else γ_{slow}
$\alpha_{t+1} \leftarrow \text{clip}_{[0,1]}(\alpha_t + \gamma_t(\alpha - e_t))$; append s_t to B
if detector is armed **then**
  Convert the new score to the standardized increment z_t
  Update $S^+_t = (S^+_{t-1} + z_t - k)_+$ and $S^-_t = (S^-_{t-1} - z_t - k)_+$
  Reset r^+_t (resp. r^-_t) to zero if the corresponding S^+_t (resp. S^-_t) hits zero; otherwise increment it
  if $\max(S^+_t, S^-_t) \geq h$ **then**
    Set $\widehat{\delta} \leftarrow r^+_t$ if the upper statistic crossed, otherwise r^-_t; record one alarm
    Keep the last $\min\{|B|, \max(n_{\min}, \widehat{\delta})\}$ scores; reset α_{t+1} to α (or the safe variant)
    Start the 100-step fast phase; disarm the detector and clear the re-anchoring list
  end if
else
  Append s_t to the re-anchoring list; after 60 new scores, set a fresh anchor, clear both CUSUM states and re-arm
end if

6 Experimental design

Data. Series of length $T = 1800$ are generated as $y_t = 10 + \ell_t + \text{trend}_t + a_t \sin(2\pi t/24) + x_t$ with x_t an AR(1) process. A single abrupt break is placed at $\tau = 1200$. Seven families are used: level shift, innovation-variance shift, added trend, seasonal-amplitude change, AR-coefficient change, plus two designed to separate detection spaces: *slow drift* (a smooth ramp of the same total size) and *predictability* (the AR coefficient falls while the innovation scale is rescaled to preserve $\text{Var}(y)$ exactly). A no-break family provides the control. Each family is run at three magnitudes and 200 seeds.

Forecaster. A ridge autoregression with 24 lags and two Fourier harmonics is fitted once on $t < 600$ and then frozen, and is applied online to the true rolling context. This captures the frozen-parameter setting while keeping the computational cost low enough for 200 replications per cell. A second arm adds instance normalization of the context window, mirroring RevIN and mean scaling. All calibration layers in a replication consume the *same* score stream, so comparisons are exactly paired.

Protocol. Calibration warm-up uses $t \in [600, 1000)$; evaluation runs over $t \in [1000, 1800)$, giving 200 pre-break and 600 post-break steps. $\alpha = 0.1$, window $W = 300$, CD horizon $H = 200$. Baselines: split conformal, rolling split conformal, ACI at $\gamma \in \{0.005, 0.05\}$, the evaluated EWA-ACI variant with five experts and exponential interval-score weights, and an evaluated Conformal PI variant with proportional and saturating integral terms but no learned scorecaster. In the released implementation these variants retain the historical code identifiers `AgACI` and `ConformalPID` for compatibility with archived CSVs; they should not be interpreted as faithful reproductions of published BOA/pinball AgACI or scorecaster-based PID. We add a **known-break-time reset benchmark** that is told τ exactly and performs BR-CP's reset with zero delay. It must still learn the new score distribution from post-break data, so it is a *known-break-time reference policy*, not a proved optimum or universal lower bound.

Inference. Coverage debt is reported with a seed-level bootstrap percentile interval (2000 resamples, vectorized as a multinomial reweighting). Per-seed paired Wilcoxon signed-rank tests compare the early post-break miscoverage rate (50 steps) and the post-break interval score, with Holm–Bonferroni correction

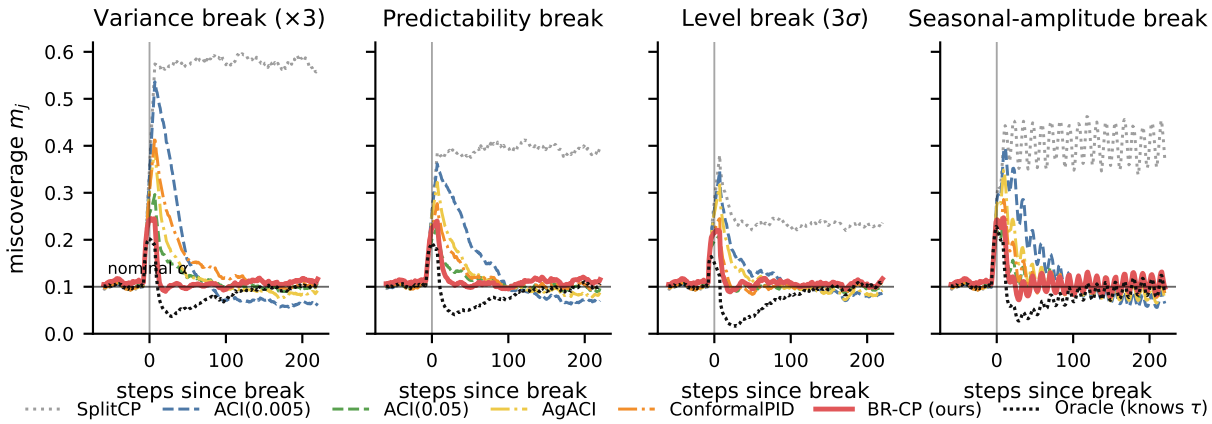


Figure 1: Instantaneous miscoverage m_j around the break (200 seeds, 15-step moving average). All methods except split conformal attain marginal coverage within 0.02 of nominal, yet they differ by up to 27× in the area above the nominal line. BR-CP (red) tracks the oracle (dotted black) closely. The seasonal panel shows residual oscillation under the evaluated unconditioned-radius rules.

Table 2: Coverage debt CD_ε at break magnitude 3 ($\varepsilon = 0.042$, $H = 200$, 200 seeds). Lower is better. The final column is the mean over the seven break families; NONE is the no-break control and quantifies the residual noise floor. The oracle is told τ; its NONE entry is an artifact of forcing a reset where there is no break.

Method	level	variance	trend	season	AR	predict.	drift	NONE	**mean**
Split CP	20.67	87.15	134.69	53.55	23.09	50.13	3.58	0.02	53.27
Rolling CP	7.95	36.11	110.10	26.30	9.77	22.13	1.52	0.00	30.55
ACI (γ=0.005)	4.15	12.02	14.69	11.46	3.55	8.58	0.51	0.02	7.85
ACI (γ=0.05)	1.15	2.81	9.05	5.76	0.35	1.60	0.09	0.02	2.97
EWA-ACI (local)	3.10	4.93	6.95	7.60	1.66	4.20	0.59	0.02	4.15
Conformal PI variant	1.62	7.43	17.67	6.85	0.66	2.87	0.14	0.01	5.32
BR-CP (ours)	**1.41**	**1.85**	**2.72**	**5.58**	**0.41**	**1.62**	0.42	0.13	**2.00**
Oracle (knows τ)	*0.76*	*1.22*	*2.98*	*3.06*	*0.86*	*1.04*	*0.84*	*0.78*	*1.54*

across the family of 21 configurations per baseline.

7 Results

7.1 Marginal coverage masks post-break dynamics

Table 2 and Figure 1 make the central point. Every Several adaptive methods attain aggregate coverage near 0.90, but averaging across configurations can mask large deficits, especially under persistent trend breaks. For example, at trend magnitude 3, BR-CP covers 0.8723 of evaluation points and ACI(0.05) covers 0.8322, both far below nominal. At that configuration, their CD values are 2.72 and 9.05, respectively; the performance difference cannot be summarized solely by a nominal-coverage statement. Their coverage debts span 0.41 to 134.69. Split conformal, which is what one gets by calibrating once on historical data, under-covers by more than 30 points for hundreds of steps and never recovers.

7.2 Comparison with the oracle

BR-CP attains a mean debt of 2.00 against 2.97 for the best tuned fixed step, 4.15 for the EWA-ACI variant and 5.32 for the PI variant, reducing the observed gap to the known-break-time reset policy (1.54) by 68%. It is numerically best among causal methods for variance, trend, and seasonal breaks at magnitude 3; faster

Table 3: Post-break interval (Winkler) score at magnitude 3, in percent relative to BR-CP. Positive means worse than BR-CP. Split conformal is omitted from the mean; its penalty reaches 604% on the trend break.

Method	variance	predict.	level	season
Rolling CP	+9.3	+2.9	−1.2	+3.1
ACI (γ=0.005)	+2.3	+0.0	−1.6	+0.3
ACI (γ=0.05)	+2.4	+1.1	+0.8	+0.8
EWA-ACI (local)	+1.4	−0.1	−1.4	+0.1
Conformal PI variant	−1.1	−2.2	−1.0	−1.1
BR-CP	0.0	0.0	0.0	0.0

ACI has lower CD in level, AR, predictability, and drift. Most of the seven-family mean benefit comes from trend: at magnitude 3, the trend CD reduction is $9.05 - 2.72 = 6.33$ of a total seven-family net reduction of approximately 6.80 (93%). Excluding trend, the reported rounded values give a mean of 1.96 for ACI(0.05) and 1.88 for BR-CP, a much smaller difference. On marginal TTR-C it recovers in 1–5 steps on level, variance, AR, predictability and seasonal breaks, against 1–25 for ACI(0.05) and 30–73 for ACI(0.005); the oracle needs 0–2.

The most practically relevant property is the scaling. As a variance break grows from 2× to 4×, BR-CP's debt moves only from 1.56 to 2.01, because larger breaks are proportionally easier to detect and the shorter delay offsets the higher per-step deficit. Over the same range The evaluated PI variant degrades from 3.16 to 11.98 and ACI(0.005) from 8.85 to 13.18. This favorable variance-break pattern is limited to the tested magnitudes and should not be extrapolated to arbitrarily large shocks; the persistent trend family is a counterexample to a blanket robustness claim.

Paired Wilcoxon tests with Holm correction find BR-CP significantly better on early post-break miscoverage in 17 of 21 configurations against ACI(0.005), the EWA-ACI variant, rolling and split conformal, and 13 of 21 against the PI variant. Against ACI(0.05) the comparison is 9 of 21 in BR-CP's favor with the reverse differences never exceeding 0.008 in absolute value. We want to be explicit about what this does and does not show. On transient coverage BR-CP and a well-tuned fast ACI are broadly comparable, and the 2.00 versus 2.97 gap in mean debt is driven by the families where a fixed fast step does badly (trend, season) rather than by a uniform advantage. What separates the two is not the transient but the frontier: ACI(0.05) and BR-CP trade lower debt against different observed volatility and width patterns (Figure 2, right), and $\gamma = 0.05$ was selected post hoc from the evaluated grid. Fair prospective comparisons require comparable development-data tuning for every method.

7.3 Interval efficiency

Post-break interval scores place BR-CP within 3% of every adaptive baseline (Table 3), and its mean interval width is *narrower* than ACI(0.005) and ACI(0.05) on six of seven families, by up to 4.6%. Because the widening is triggered rather than continuous, the intervals are wide when the regime is unknown and tight otherwise, which is the behavior the interval score rewards.

7.4 Fixed-step sensitivity

Figure 2 sweeps γ over three orders of magnitude on the variance break. Debt falls from 26.34 at $\gamma = 0.001$ to a minimum of 2.81 at $\gamma = 0.05$ and then rises again to 10.47 at $\gamma = 0.2$, while steady-state coverage on the no-break control degrades monotonically from 0.9025 to 0.8802. BR-CP attains 1.85 at a steady-state coverage volatility of 0.0375, which on the ACI curve is larger than the volatility of ACI($\gamma = 0.05$), even though BR-CP has lower CD. The plotted no-break coverage volatility is 0.0375 for BR-CP and 0.0262 for ACI($\gamma = 0.05$), so BR-CP pays a measurable volatility cost; the figure does not establish dominance at both coordinates. Volatility here is the reported standard deviation of empirical coverage-rate estimates across no-break replications, not the step-to-step variation of interval radius or of α_t.

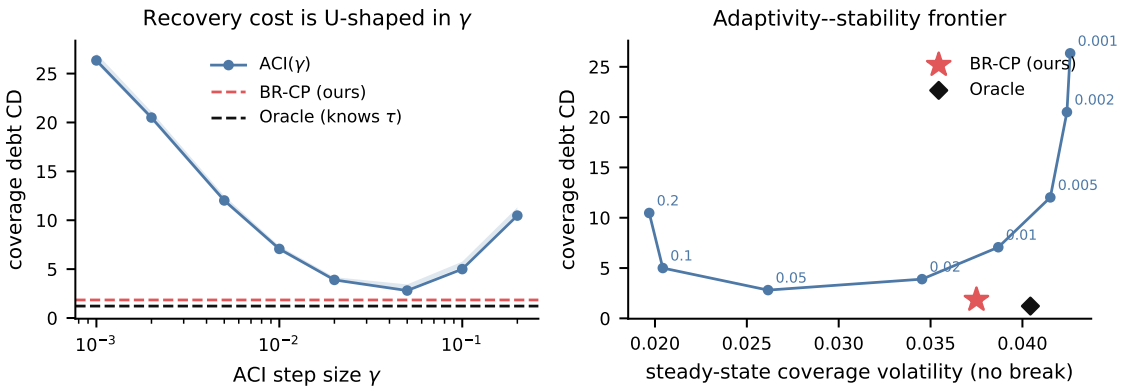


Figure 2: Left: coverage debt of ACI as a function of γ is U-shaped; the best attainable value, 2.81 at $\gamma = 0.05$, still exceeds BR-CP's 1.85. Right: coverage-debt versus variability in empirical no-break coverage. BR-CP has lower CD than tested ACI steps, but not lower volatility at every operating point.

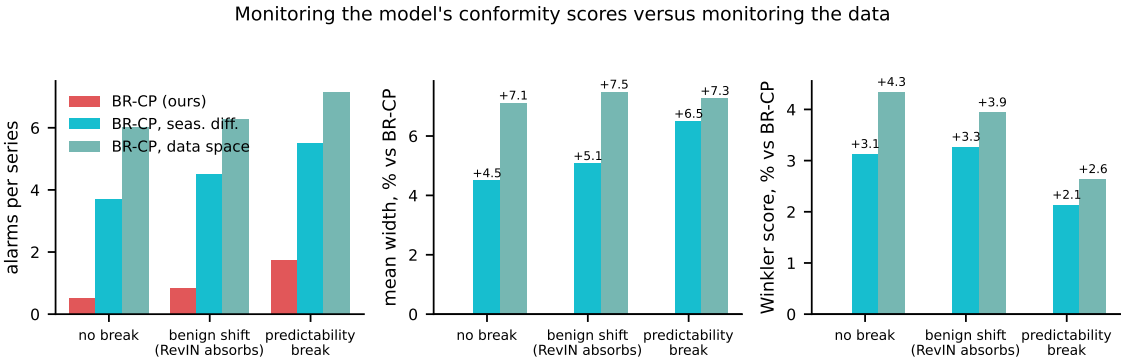


Figure 3: Score-space versus data-space monitoring, identical CUSUM recursion. At the tested, unmatched false-alarm operating points, monitoring observations produces more alarms and pays 4.5–7.5% in interval width and 2.1–4.3% in interval score. On a stationary seasonal series with *no break at all* the data-space monitor fires six times per series.

7.5 Detector-space comparison

Figure 3 compares the identical detector applied to the score stream, to the raw observations, and to seasonal differences, the strongest model-free way to strip predictable structure before monitoring in data space. On a stationary series with no break, BR-CP raises 0.52 alarms per 800 steps; the data-space variant raises 6.00 and the seasonally differenced variant 3.70. Under a benign level shift absorbed by instance normalization the counts are 0.82, 6.27 and 4.49.

The consequence is not a coverage failure, since all three variants remain valid and the trigger-happy ones are if anything slightly conservative, but a systematic efficiency loss: 4.5–7.5% wider intervals and 2.1–4.3% worse interval scores across all three regimes. These findings compare the implemented operating points. They do not isolate the effect of monitored input because the false-alarm rates differ; a separately tuned, matched-ARL experiment is needed. One might object that a better data-space monitor should first remove predictable structure; the seasonal-difference variant does exactly that, and narrows but does not close the gap. Taken to its conclusion, the objection is the argument: removing predictable structure requires a model, and monitoring the residuals of that model *is* score-space monitoring.

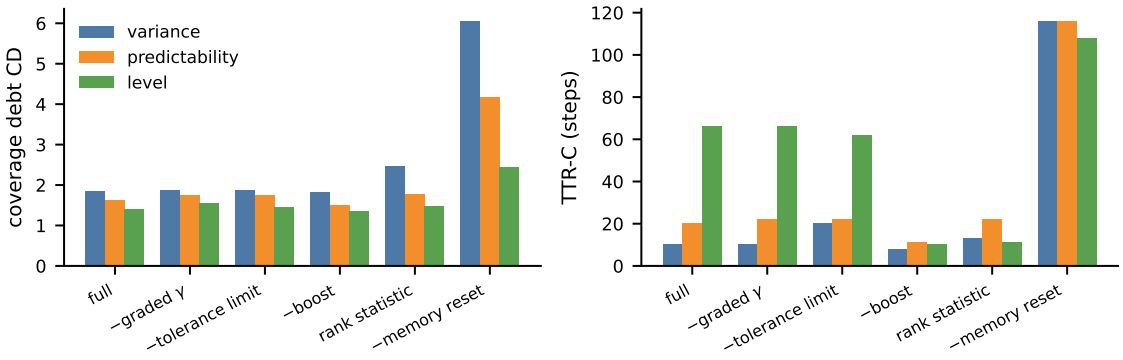


Figure 4: Contribution of each mechanism. Removing the memory reset is the most damaging; the rank-based detector statistic is uniformly slower than the standardized one.

7.6 Ablation

In these tested ablations, removing the memory reset (R2) is especially costly: debt rises from 1.85 to 6.05 on the variance break and from 1.62 to 4.16 on the predictability break, and TTR-C from 10 and 20 steps to 116. The graded step size (R1) contributes a smaller but consistent gain (1.55 → 1.41 on level, 1.73 → 1.62 on predictability) at a small steady-state cost. Replacing the standardized detector statistic by the rank-based statistic raises debt to 2.46 on the variance break and widens intervals by 2.0%, showing a longer delay for the tested rank-based operating point; a numerical rank cap alone cannot explain the result without accounting for threshold and slack normalization.

The 100-step post-alarm boost (an optional part of R1, not R4) is *not* supported as an additional improvement: removing it leaves debt statistically unchanged or marginally better (1.81 vs. 1.85 on variance). Once the graded step size and the reset are in place the boost is redundant, and we report this rather than retaining a component the evidence does not justify. A no-boost design may be preferable in deployment, but the main estimates in this paper still refer to the evaluated boosted implementation.

7.7 Mechanism illustration

Figure 5 traces one replication and makes concrete what the aggregate numbers describe. The CUSUM statistic is near zero before the break, so the step size sits at γ_{slow} and the layer behaves like conservative ACI. Two steps after the break the statistic crosses its threshold; the memory reset discards the pre-break scores and the tolerance limit, evaluated on the handful of post-break scores that triggered the alarm, raises the radius from 1.6 to about 5 in a single step. ACI(0.005) reaches the same radius after roughly fifty steps. The visible jaggedness of the BR-CP radius over the following hundred steps is the tolerance limit at work on a buffer that is still small; as the buffer refills the margin contracts and the radius settles onto a path slightly below ACI's, which is the width advantage of Table 3 seen at the level of a single series.

8 Controlled stress tests on real-world benchmarks

8.1 Frozen cross-domain forecaster

The simulation isolates the calibration layer, and that isolation is what makes it insufficient on its own: real conformity scores are heavier tailed, autocorrelated and heteroskedastic in ways synthetic AR noise is not. We therefore repeat the core comparison using a frozen cross-domain linear forecaster on real target series. This models the calibration aspect of deployment, but is not an experiment with a published foundation model.

The archived protocol reports fitting a single global linear forecaster once on 261 training series drawn from *electricity*, *traffic* and *weather* (313,200 context windows), then freezing the resulting weights. This revision recovered the saved weights and reproduced the target evaluation, but did not reproduce that original

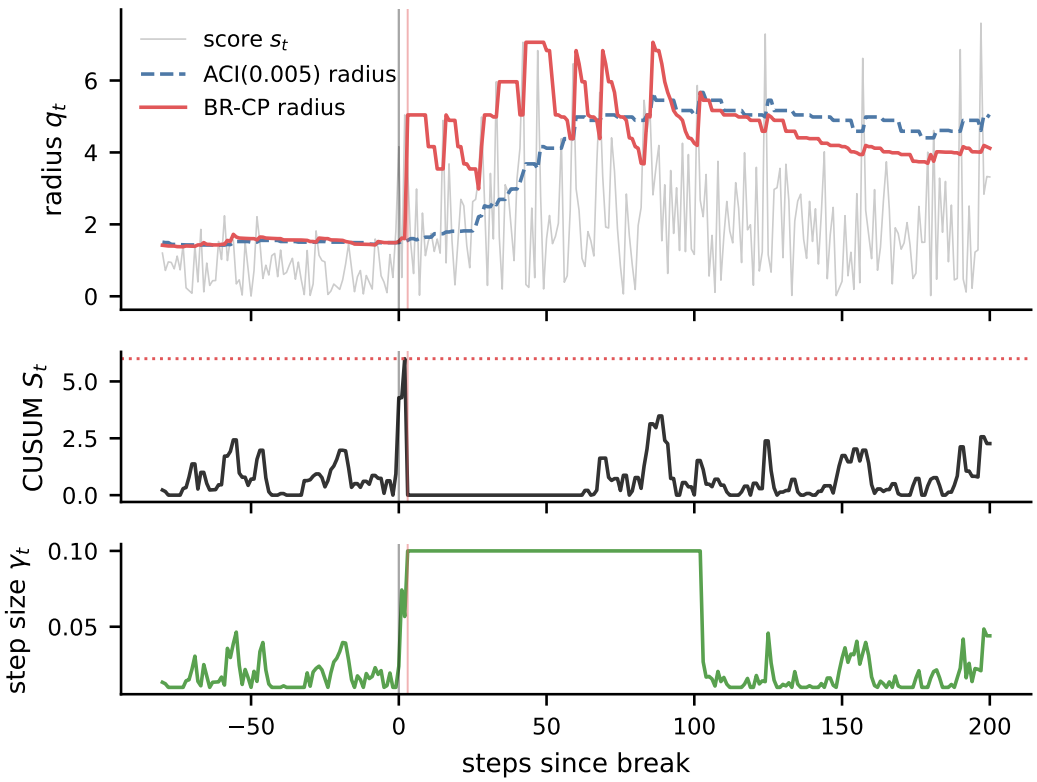


Figure 5: A single replication of the variance break. The CUSUM crosses its threshold two steps after the break; the radius jumps immediately from 1.6 to 5, where ACI(0.005) needs about fifty steps to arrive. The radius is visibly jumpy while the post-break buffer refills under the tolerance limit, then settles.

pretraining from identical source files. It regresses the next value on an instance-normalized context window of 96 observations and is frequency-agnostic. It is then applied, without retraining, to target datasets reported as excluded from its fitted corpus: *ETTh1*, *ETTh2*, *ETTm1*, *ETTm2* and *exchange rate*. Because one recovered electricity file had the wrong benchmark schema, we cannot independently certify exact training-data versions or byte-level train/evaluation exclusion. The archived report describes this as cross-domain transfer, and reports that mean absolute error is 0.31–0.51 times that of seasonal naive on every held-out dataset. (*Influenza* is excluded: at 966 weekly observations it cannot supply a calibration window, a break window and a recovery horizon.)

Coverage debt is undefined without a known break time, and the natural regime changes in these benchmarks are not dated. We therefore inject bounded, non-cumulative breaks of identical size into the *raw* target series, so the frozen model forecasts the broken series exactly as it would in deployment while τ is known exactly. Windows around every break, pooled across all series, are the replications: 322 break windows over 35 series and 368,081 calibration steps per layer.

The full target-data arms (level, variance, no break) were replayed using the archived frozen forecaster and corrected deterministic Wilks cache. Across 35 series, the variance-arm BR-CP CD remained 3.210874 and coverage moved from 0.895914 to 0.895936; BR-CP-safe CD moved from 3.148036 to 3.144931, and its coverage moved from 0.896813 to 0.896819. These small changes do not eliminate the nominal-coverage deficit or establish fidelity to unimplemented published baselines.

8.2 Detector recalibration

The threshold selected on synthetic data, $(k, h) = (0.65, 6.0)$, gives $\mathrm{ARL}_0 = 1882$ there but only 642 on real conformity scores. The simulation's limitations section anticipated this; the size of the effect is worth

Table 4: Controlled variance-shift stress test on five real-world benchmark datasets: 322 break windows over 35 held-out series. CD uses $\varepsilon = 0.03344$ for 322 pooled windows, whereas the synthetic arm uses 0.04243 for 200 seeds; TTR-C uses the 24-step marginal recovery criterion. The final column reports alarms per 1000 steps on unbroken series.

Method	CD	TTR-C	cov.	Winkler	alarms
Split CP	114.05	∞	0.658	17.64	0.00
Rolling CP	43.02	∞	0.880	10.96	0.00
ACI (γ=0.005)	16.85	55	0.901	10.07	0.00
ACI (γ=0.01)	9.10	28	0.901	9.54	0.00
ACI (γ=0.05)	3.82	27	0.895	8.93	0.00
ACI (γ=0.1)	6.15	72	0.888	8.93	0.00
EWA-ACI (local)	4.67	43	0.893	8.92	0.00
Conformal PI variant	17.44	∞	0.899	9.74	0.00
BR-CP	**3.21**	**8**	0.896	**8.91**	1.05
BR-CP-safe	**3.14**	**6**	0.897	8.93	1.05
BR-CP, data space	3.48	13	0.902	9.27	7.62

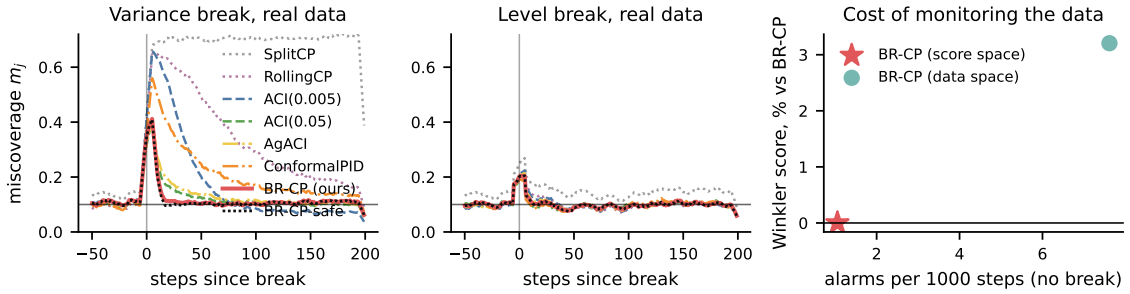


Figure 6: Controlled stress tests on real-world benchmark series. Left: post-break miscoverage after a variance shift. Middle: a 3σ level shift has little effect after instance normalization. Right: on unbroken series, data-space monitoring produces substantially more alarms than score-space monitoring and a higher interval score.

recording. Recalibrating on 75 held-out series from the corpus datasets, which are neither pretraining nor evaluation data, selects $(k, h) = (0.8, 8.0)$ with $\mathrm{ARL}_0 = 1744$. All real-data results below use that operating point. Because real series contain genuine regime changes, this is an upper bound on the false-alarm rate rather than a pure null, so the threshold it implies is conservative.

8.3 Transfer results

Table 4 and Figure 6 show the same method ranking for the variance-shift stress test. On the variance break BR-CP attains the lowest coverage debt, 3.21 against 3.82 for the best evaluated ACI, 4.67 for the EWA-ACI variant and 17.44 for Conformal PID, and crosses the tolerance-adjusted 24-step moving-average recovery threshold in 8 steps, versus 27 to 55 for selected adaptive baselines; the evaluated PI variant is censored beyond the reported horizon. Early post-break miscoverage is 0.178 against 0.205 for ACI(0.05) and 0.387 for the PI variant. Its Winkler interval score is 8.91, compared with 8.93 for ACI(0.05) and 8.92 for EWA-ACI.

Two further predictions transfer. First, monitoring the observations rather than the conformity scores costs what the simulation said it would: on unbroken real series the data-space variant raises 7.62 alarms per 1000 steps against BR-CP's 1.05, for 5.5% wider intervals and a 3.2% worse interval score at the evaluated operating points, which are not matched for no-break false-alarm rate. Second, and more strikingly, the *level* break is almost benign on real data: every adaptive method scores a coverage debt near 1 and recovers within a step or two, consistent with attenuation of a level shift under the evaluated instance normalization. This controlled result, rather than the dated-event outcomes, supports the narrower interpretation that calibration should track model competence rather than the data.

8.4 Failure cases and attenuated effects

We report three results that qualify the claim.

The margin over a tuned fast ACI shrinks. In simulation BR-CP beat ACI(0.05) by 34% on mean debt; here the gap is 16% (3.21 vs 3.82), and a paired Wilcoxon test across the 35 series finds *no* significant difference in interval score (-0.14%, $p = 0.49$), nor against the EWA-ACI variant (-0.30%, $p = 0.37$). The differences that are significant are against ACI(0.005) ($+12.2\%$, $p < 10^{-4}$), ACI(0.01) ($+6.4\%$), the PI variant ($+8.2\%$), rolling conformal ($+22.0\%$) and the data-space variant ($+3.2\%$). The defensible claim on real data is therefore that BR-CP has a small observed interval-score difference relative to the evaluated fast ACI and EWA-ACI variants; neither of the reported Wilcoxon comparisons detected a significant difference, while the tolerance-adjusted recovery estimates differed. Nonsignificance does not establish statistical equivalence.

The safe reset slightly improves the perturbed-data point estimate. BR-CP's marginal coverage on real data is 0.896; the safe-reset variant, which avoids increasing the internal miscoverage level on alarm (without guaranteeing a monotone radius), reaches 0.897 with a slightly *lower* debt (3.14 in the corrected-cache replay; 3.15 historically) and faster recovery (6 steps) at a 0.8% width cost. A separate unbroken-series setting yields 0.900. A newly calculated 35-series cluster bootstrap on the archived perturbed variance-test per-series summary gives BR-CP coverage 0.895914 (95% percentile interval $[0.894722, 0.897162]$), safe coverage 0.896813 ($[0.895624, 0.898049]$), and paired safe-minus-standard difference 0.000899 ($[0.000657, 0.001144]$). These intervals resample whole series, not individual prediction steps, and do not necessarily capture dependence between series from the same benchmark. Neither result establishes that the method removes the nominal coverage deficit under injected breaks or in the simulations. The variant is a pragmatic alternative rather than a coverage-guaranteed default.

Closely spaced breaks defeat the re-anchoring rule. When a second break of a different kind is injected 40 steps after the first, inside the 60-step window during which the detector is disarmed and rebuilding its null, BR-CP's debt rises to 4.28, *worse* than ACI(0.05)'s 3.87, although recovery is still faster (15 versus 20 steps). The safe variant recovers part of it (4.18, 13 steps). This is a real failure mode of mechanism R4 and not a tuning artifact: while re-anchoring, the layer is blind. Maintaining a shadow anchor from the pre-alarm regime could reduce this blind period and is left for future work.

Threshold sensitivity. Varying h by $\pm 30\%$ around the calibrated value moves debt from 2.72 ($h = 5.6$, 38.1 alarms per series, or 3.62 per 1000 steps) to 3.21 ($h = 8.0$, 27.3 per series, or 2.60 per 1000 steps) to 3.48 ($h = 10.4$, 23.1 per series, or 2.20 per 1000 steps), with marginal coverage between 0.894 and 0.897 and mean width varying by under 1%. Across this sensitivity range, the main coverage and width conclusions remain similar. The most conservative threshold has higher debt than the calibrated setting, whereas the lower threshold trades more alarms for lower debt.

8.5 Externally dated events

The controlled real-data study injects breaks to obtain known break times and replicated miscoverage curves. As a complementary descriptive analysis, we replayed two externally dated events using the recovered `exchange_rate.csv` and `national_illness.csv` inputs, the archived frozen global ridge forecaster, and the same calibration implementations. The event dates are 15 September 2008 (Lehman Brothers bankruptcy) and 26 April 2009 (US H1N1 emergency announcement). The first weekly influenza observation on or after the latter date is 28 April 2009. These dates are external anchors, *not* verified statistical change points. There is one observed realization per series, so we report realized miscoverage, rather than population conditional coverage or a coverage-debt estimand.

The recovered run reproduces all 84 method-by-series rows from the archived case-study CSV exactly, including full-period coverage, pre/post miscoverage, width, and interval scores. The separate local trace export contains 32,480 time-indexed records and allows us to correct a bookkeeping error in the archived `alarm_lag` column: the original script subtracted the warm-up length a second time even though the alarm index was already relative to the evaluation window. Table 5 and Figure 7 report the corrected time origin and local observed hit-rate trajectories. Method means in the table average equally across the eight exchange-rate or four illness channels; they are not independent replications of a break-probability distribution.

No BR-CP alarm occurs at offset zero for either event. During the 260 post-Lehman daily observations, however, the score monitor raises seven alarms over five of the eight channels. The earliest post-event alarm offsets for those five channels are $11, 31, 48, 96, 257$ observations; the remaining three channels have no

Table 5: Observed pre/post miscoverage for the externally dated events. The Lehman window contains 260 daily observations after the event, and the H1N1 window contains 60 weekly observations. Post-event alarm counts are from BR-CP's own CUSUM and are summed over channels; *none* is registered at offset zero. These are descriptive rates, not conditional coverage estimates.

Event	Calibration method	Pre miscov.	Post miscov.	Channels	Post alarms
Lehman	SplitCP	0.0942	0.1808	8	–
Lehman	ACI(0.05)	0.1115	0.0971	8	–
Lehman	BR-CP	0.1000	0.0962	8	7
Lehman	BR-CP-safe	0.0990	0.0947	8	7
H1N1	SplitCP	0.3083	0.2000	4	–
H1N1	ACI(0.05)	0.1250	0.0917	4	–
H1N1	BR-CP	0.1125	0.0917	4	0
H1N1	BR-CP-safe	0.1125	0.0833	4	0

post-event alarm in the window. Thus the previous statement that the detector produced *no response* to Lehman was false: it produced delayed and channel-specific responses, but no same-day alarm. During the 60-week post-H1N1 window, it raises no alarms in the four illness channels. Alarm-free windows do not imply the absence of local score variability: the empirical 25-step hit rates vary visibly near both event dates. For Lehman, the mean observed BR-CP miscoverage changes from 0.1000 to 0.0962 across the pre/post windows, whereas static split conformal changes from 0.0942 to 0.1808. For H1N1, BR-CP changes from 0.1125 to 0.0917; neither pooled average establishes a population-level conditional statement.

The source CSVs have the expected columns and date ranges and have been hashed in the accompanying replication audit. Their exact byte identity to the upstream public release remains unverified. The replacement `electricity.csv` found elsewhere on the connected Drive is *not* the THUML Electricity benchmark and is excluded from any claim of newly retrained pretraining; the event study instead uses the previously archived frozen forecaster, unchanged. These considerations limit external validity, and no causal claim is made for the effectiveness or failure of detection at the external event dates.

Real variance-arm mechanism audit. The new full 35-series variance-arm replay was independently instrumented for the published implementations ACI(0.05) and BR-CP. Both outputs match the deterministic corrected-cache full replay in all coverage, width, interval-score, and total-alarm fields for the 35 paired series. Across 368,081 evaluated forecasts per method, ACI(0.05) projects its internal level at 4,463 updates (1.213%), evaluates a nonpositive level at 4,315 radii, and caps 803 further finite empirical quantiles at the sample maximum. For BR-CP, the corresponding update projections number 1,720 (0.467%); nonpositive-level radii number 952, and there are 66 finite empirical-quantile caps. The BR-CP implementation enters the Wilks-inspired small-buffer order-statistic calculation 210,648 times and uses the empirical maximum because no order statistic meets the nominal fixed-i.i.d. criterion on 6,247 occasions (2.966%). Neither of these two methods emits an infinite radius in this replay. These are implementation frequencies, not validity guarantees, and should not be extrapolated to published AgACI/PID or other stress-test arms. The per-series count CSV and an instrumented script are included in the revision audit as separately versioned evidence.

8.6 Remaining validation

Our forecaster is a small global model, not a published checkpoint. The protocol above is checkpoint-agnostic: `scripts/run_tsfm_real.py` implements the same interface for Chronos-Bolt and TimesFM. The detector threshold is likely to require re-estimation because the conformity-score dependence structure changes with the forecaster and dataset. Confirmatory evaluation on published checkpoints and multi-horizon calibration remain future work.

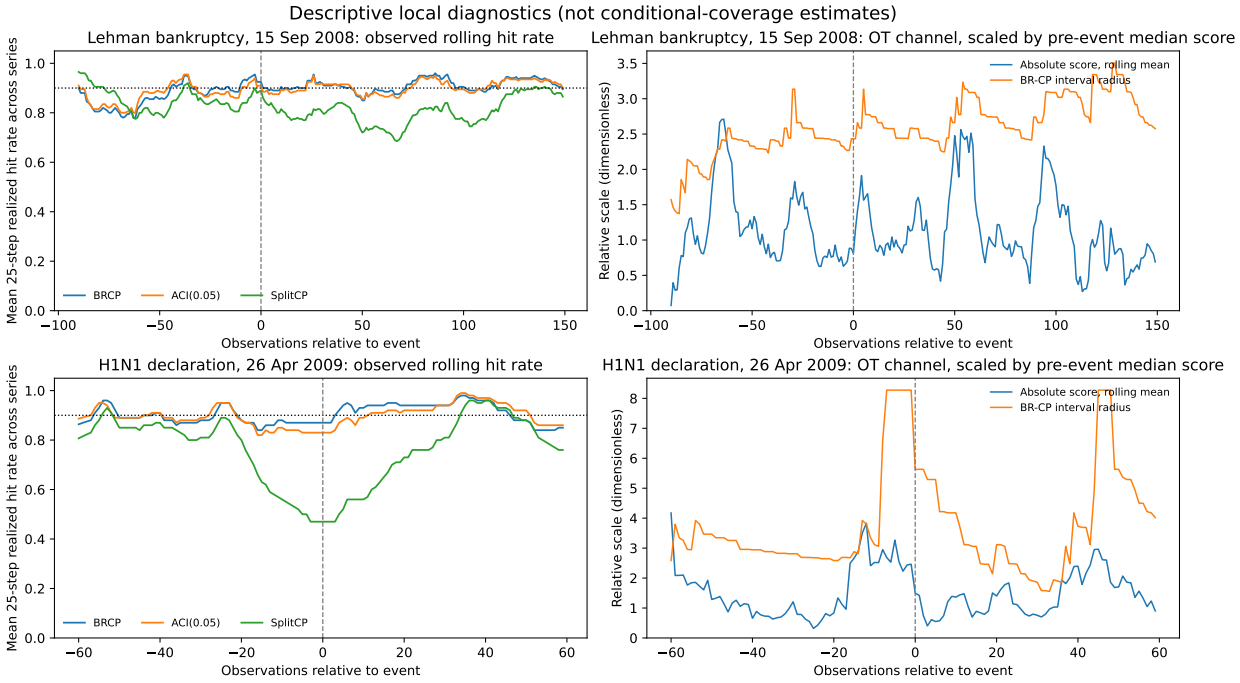


Figure 7: Replayed, descriptive local diagnostics around the two event dates. Left panels show mean realized 25-step trailing hit rate over the respective channels; right panels show the `OT` channel's absolute residual score (rolling mean) and BR-CP radius, both normalized by its pre-event median score. A vertical dashed line marks the prespecified external date. These curves are not estimates of conditional coverage and do not validate a causal event-induced break.

Table 6: Mean synthetic CD over 7 families, 3 magnitudes, 200 paired seeds per cell. Results recalculated from released per-offset error arrays.

ε	BR-CP	ACI(0.05)	EWA-ACI (local)	PI variant
0.000	5.450	6.351	7.986	9.311
0.020	3.230	4.285	5.686	7.046
0.04243	1.915	2.828	4.001	5.356
0.050	1.651	2.468	3.591	4.903
0.070	1.203	1.712	2.748	3.870

9 Additional uncertainty and mechanism checks

Tolerance sensitivity. We recomputed CD directly from archived per-replication error curves, without running new forecasts. Table 6 averages the 21 synthetic break cells, and includes $\varepsilon = 0$. The method ranking shown here persists over the examined grid, but this is an empirical observation, not a mathematical invariance of rankings.

Real-data dependence and weighting. The 322 stress-test windows are nested within 35 held-out series. In Table 4, every window receives equal weight; thus series contribute in proportion to their number of windows. To assess sensitivity, we resampled *entire series* with replacement (2,500 draws), retaining all windows within each resampled series and recomputing the nonlinear CD estimator. The paired bootstrap distribution for $\mathrm{CD}_{\mathrm{ACI}(0.05)} - \mathrm{CD}_{\mathrm{BRCP}}$ has a percentile range of $[0.261, 1.017]$ with observed difference 0.611. The same 35-series resampling yields marginal TTR-C of 8 steps (percentile range $[7, 9]$) for BR-CP and 27 ($[25, 32]$) for ACI(0.05). These are uncertainty intervals for the operational period-average recovery estimator, not intervals for an unobserved population individual-series recovery time. Equal weighting of the 35 series

Table 7: New synthetic ACI boundary-convention audit, 60 independent seeds per row and 48,000 evaluated steps per row. "$+\infty$" counts emitted infinite radii; efficiency comparisons for that convention are not finite.

Break	γ	clipped cov.	unprojected cov.	clipped projections	$+\infty$ radii
No break	0.05	0.8995	0.8998	89	397
No break	0.20	0.8810	0.8998	2,690	10,323
Variance	0.05	0.8941	0.9000	705	2,473
Variance	0.20	0.8626	0.8998	4,182	14,409
Trend	0.05	0.8320	0.8975	6,236	18,743
Trend	0.20	0.7336	0.8987	11,947	29,413

instead yields BR-CP CD 3.251 and ACI(0.05) CD 3.788, compared with window-weighted CD 3.211 and 3.822. This cluster procedure accounts for within-series windows, but not necessarily dependence across series from the same benchmark. Since CD applies a positive part to estimated probabilities, bootstrap intervals for individual method CD can be shifted upward; paired *differences* are emphasized rather than declaring strict coverage of the observed plug-in point estimate by marginal intervals.

On the real variance-stress-test curves, $\varepsilon = 0$ gives CD 5.532 for BR-CP versus 6.780 for ACI(0.05); the reported main-data tolerance 0.03344 gives 3.211 versus 3.822, while 0.042 gives 3.043 versus 3.463. The tolerance-adjusted 24-step marginal TTR-C is 8 versus 27 at the main tolerance and 64 versus 130 at $\varepsilon = 0$. Recovery estimates beyond the horizon are censored, not assigned a true recovery time of 201.

Detector-plus-reset-only ablation. We added an explicit implementation-level ablation consisting of the same score-CUSUM, its detected reset-location rule, and fixed-step clipped ACI ($\gamma = 0.01$), but *without* graded adaptation, the 100-step fast phase, or the small-buffer tolerance adjustment. This is an internal *detector-plus-reset-only* reference and is not presented as a reproduction of any published CUSUM-reset method. Across the 21 synthetic break cells, mean CD was 2.406 for this baseline versus 1.915 for the main BR-CP. The improvement is heterogeneous: at magnitude 3, trend CD was 6.981 versus 2.716, but seasonal CD was 4.998 versus 5.581, and slow-drift CD was 0.206 versus 0.425. These comparisons help isolate when the additional mechanisms help and when they cost debt.

9.1 Boundary conventions, detector-matching attempt, and reset-buffer audit

ACI boundary sensitivity (new independent synthetic replay). The practical implementations in the main tables project the internal ACI miscoverage level onto $[0, 1]$ and cap unresolved upper quantiles at the finite buffer maximum. To quantify the effect of this choice, we independently replayed 60 newly generated seeds per condition (seeds 40000–40059), using the same frozen-forecaster generator, under a second convention: the *unprojected* ACI recursion, for which empirical quantiles outside the unit probability range yield infinite or empty sets. The latter convention preserves the unprojected coverage-control logic but sometimes makes interval width and Winkler score infinite or undefined. These are new, separately seeded checks and do not replace the original simulation grid. For example, at trend magnitude 3 with $\gamma = 0.05$, using the unprojected boundary raises whole-period coverage from 0.8320 to 0.8975, but 39.0% of emitted intervals have infinite radius. The original ACI rule cannot therefore be compared with the capped finite-radius implementation on mean finite width without explicitly reporting this cost. The new audit script and per-seed results are included in Online Resource 1.

Development-targeted false-alarm matching (new synthetic replication). To improve comparability across detector spaces, we separately selected the threshold for score-space and raw-data-space Page-CUSUM on 50 independent no-break development streams (seeds 10000–10049). We held $k = 0.65$, the standardized statistic, the 250-point anchor and 60-score re-arm rule fixed. On a fixed threshold grid, the targets were 0.5 alarms per 800 no-break evaluation steps; the selected thresholds were $h = 5.75$ (score) and $h = 10.50$ (data), producing 0.46 and 0.50 alarms per 800 *development* steps. These settings were not reselected on test data. In the independent test null series, score and data monitors raised 0.517 and 0.833 alarms per 800 steps, respectively; a paired seed bootstrap gives a 95% percentile interval of approximately $[0.00, 0.65]$ for

Table 8: Held-out synthetic detector-space replay (60 paired test seeds, 30000–30059). CD uses 200 offsets and $\varepsilon = 0.04243$. Development false-alarm targets were similar, but the held-out null rates diverged; this is not a perfectly matched-ARL experiment.

	No-break alarms/800		Variance CD		Predictability CD	
Detector	development	test	test	test TTR-C	test	test TTR-C
Score, $h = 5.75$	0.46	0.517	2.517	4	2.444	8
Data, $h = 10.50$	0.50	0.833	3.564	10	2.085	16

Table 9: New instrumentation for the complete historical synthetic main grid (200 replications at each of three magnitudes per family; $h = 6.0$). The final column counts projected BR-CP calibration updates, not ACI updates.

Family	R3 calls	Max fallback	Projected updates
AR	298,044	18,140	2,469
Level	191,471	4,073	463
Predictability	233,625	7,336	856
Season	199,639	6,782	792
Slow drift	196,250	4,333	510
Trend	336,835	98,031	17,777
Variance	223,128	8,646	943
Total	1,678,992	147,341	23,810

the *data-minus-score* alarm-count difference. The difference in operating points persists despite calibration on development data. The data monitor also had lower CD in the predictability setting but slower event detection conditional on alarm, so the results do not establish universal superiority of score-space monitoring. We retain the previously reported comparisons as *unmatched* and do not draw an isolated causal monitoring-space conclusion from either study. More development samples, direct ARL_0 matching and additional held-out real series are needed.

R3 fallback and ACI projection counts (new synthetic subset). We instrumented all BR-CP radius evaluations in the 60 held-out test seeds across the no-break, level, variance, season and predictability arms, with the *new matched-development score threshold* $h = 5.75$, not the main-table $h = 6.0$ setting. Across 240,000 emitted score-space intervals, there were 347 projected-level updates and 233 finite-maximum quantile caps (including 226 evaluations at $\alpha_t \leq 0$). The R3 small-buffer order statistic was called 98,022 times; on 3,240 of these calls (3.31%), no order statistic met the nominal 90% Wilks confidence criterion and the code used the sample maximum. These event counts are direct code instrumentation under the specified exploratory-threshold protocol; they are not extrapolated to the original 21-cell main simulation or to the real-data arm. The maximum fallback is *not* certified at 90% confidence for its selected, dependent buffer.

Full 21-cell historical-configuration replay (new confirmed synthetic audit). We additionally instrumented all seven break families at all three magnitudes, using the original seeds 0–199 and the original main operating point $k = 0.65, h = 6.0$. Crucially, this historical-configuration replay used the same ordered BR-CP-then-OracleCP evaluation as the released program: the original tolerance-order-statistic cache was process-global, so running the methods in a different order can change later quantiles. The replay reproduced the archived *per-seed* BR-CP coverage exactly and width to a maximum absolute numerical discrepancy of 7×10^{-12}. Across 3,360,000 emitted BR-CP intervals, the R3 rule was invoked 1,678,992 times; 147,341 calls (8.78%) fell back to the sample maximum. The BR-CP adaptive level was projected back onto $[0, 1]$ after 23,810 updates (0.709% of steps); 20,509 subsequent radius evaluations occurred at $\alpha_t \leq 0$, and 211 other unresolved finite quantile levels were capped at the sample maximum. No infinite-radius intervals were emitted. The fallback is not an i.i.d. tolerance certificate.

Full original clipped-ACI baseline audit. For the two fixed-step ACI baselines, we separately replayed the complete 21×200 original synthetic configurations and checked every seed against the archived coverage and width outputs (agreement at numerical precision). For ACI(0.005), 4,484/3,360,000 updates (0.133%) were projected onto $[0, 1]$; 4,647 radii were evaluated at $\alpha_t \leq 0$ and 38,337 additional finite-quantile evaluations were capped at the sample maximum. For ACI(0.05), the corresponding counts were 88,828/3,360,000 (2.644%), 88,719, and 8,207. Neither clipped/capped ACI implementation emitted infinite radii. These quantities concern the main *modified* ACI baselines, not the separate unprojected ACI sensitivity study in Table 7.

Deterministic cache correction and sensitivity. The archived implementation keyed the Wilks-order-statistic memoization on α_t rounded to three decimal places, but calculated the initial cached answer using the unrounded level. Consequently, different ordering of BR-CP and OracleCP could produce different later interval radii. We corrected the numerical cache in a separately versioned source fork: the lookup key and order-statistic calculation now use the same level rounded to twelve decimal places. An independent full 21-cell, 200-seed replay with this fix changed cell-average BR-CP coverage by at most 3.75×10^{-5} and cell-average mean width by at most 0.00104 in absolute value. BR-CP mean CD over the 21 break cells changed from 1.91538 to 1.91467; the magnitude-3 mean changes from 2.00117 to 2.00154, leaving the conclusions at the reported two-decimal precision unchanged. The original archive, corrected implementation, per-seed comparisons, and provenance are supplied separately so the numerical correction is not hidden.

10 Post-hoc published-checkpoint robustness audit

We additionally performed a *post-hoc, exploratory* one-step experiment with the frozen public Chronos-Bolt Small checkpoint (revision `772f3d25d38aec6d914c8949dab4462e2d46f5d8`). This is *not* the source of the historical numerical claims in Sections 6–8, and it does not establish training-corpus disjointness or strict zero-shot evaluation. It uses six public evaluation datasets (ETTh1, ETTh2, ETTm1, ETTm2, exchange rate, illness), six evaluated channels per dataset (36 in total), three scenarios per channel (unperturbed, level pulse, variance pulse), and nine locally implemented calibration methods (108 scenario streams, 972 method–stream records). The original Chronos study classifies ETT (hourly and 15-minute variants) and Exchange Rate among the 27 *zero-shot evaluation* benchmark families excluded from its training sets (Ansari et al., 2024, Table 3). The Chronos-Bolt model card describes zero-shot testing on the corresponding benchmark suite. These family-level documents support benchmark-intent separation for five of the six named datasets; they do not verify byte-level identity of our THUML files with those benchmark versions, and they do not certify the training membership of national illness. We therefore report these as public benchmark-provenance evidence, not a verified six-file pretraining-disjointness guarantee. Detector thresholds in this post-hoc arm were selected using one held-out channel per dataset, with remaining channels reserved for evaluation. These channels are disjoint but not statistically independent: they share dataset-level temporal structure, so the test is not a new-domain, independent hold-out. Each method uses the *same cached* Chronos median forecasts for its corresponding stream, enabling a paired comparison. A level or variance intervention lasts 120 time steps and then returns to baseline; this engineered return is a second change.

Corrections before analysis. Our first GPU notebook had two defects. It passed the conformal rank $k = \lceil (n+1)(1-\alpha) \rceil$ as probability k/n into a "higher" interpolation quantile, selecting the next order statistic when $k < n$. We replaced this with the direct one-based rank k, retaining the documented finite sample-maximum fallback when $k > n$; the latter does not prove finite-sample validity under adaptive, dependent scores. We replayed all 972 records from the original saved forecast traces without running inference again, and added 114 order-statistic tests. The original detector search over $h \leq 20$ was also too narrow to match data-space and score-space alarm frequencies. A *post-hoc expanded development grid* selected $h_{\text{score}} = 10$ and $h_{\text{data}} = 500$; on unperturbed evaluation streams the observed rates were respectively 0.4861 and 0.5637 alarms per 800 steps (difference 0.0777 against the numerical tolerance 0.15). This is matching of *empirical* alarm counts, not an independently confirmed equality of null false-alarm rates. The grid was expanded after the previous test mismatch became known, so this analysis is not held-out confirmation.

Recovery estimates without the return break. Because the injected pulse ends at offset 120, computing coverage debt over 150 offsets would confound the original intervention with the engineered return. We therefore report *exploratory* $\mathrm{CD}_{\varepsilon,100}$ and TTR-C on offsets $j = 0, \ldots, 99$, each before the return. At each offset, we average the binary errors across repeated injected events *within* a channel and then average across the 36 channels. For the 24-step rolling-average first-crossing TTR-C, the earliest $j \leq 76$ satisfying $24^{-1} \sum_{r=j}^{j+23} \widehat{m}_r \leq \alpha + \varepsilon$ is reported; if none exists, TTR-C is right-censored. A separate strict measure requires 25 consecutive *marginal-offset* values below the threshold and is censored unless such a sequence occurs by offset 75. Neither statistic is a conditional-coverage guarantee, and first crossing need not be persistent. We illustrate $\varepsilon = 0.03344$, which is a sensitivity point reused from the historical real-data analysis, *not* a newly validated confidence bound for these 36 related channels.

Table 10: Exploratory post-correction Chronos-Bolt Small audit. CD100 and TTR24 are computed before the offset-120 return break, with $\alpha = 0.10$, $\varepsilon = 0.03344$. TTR24 is the first 24-step rolling-average crossing; strict25 is a different condition. "> 75" denotes right censoring of strict25 in the 100-step window.

Pulse	Wrapper	Coverage debt 100	TTR24	Strict25	Mean miscoverage
Level	ACI(0.05)	1.398	1	17	0.107
Level	BR-CP	1.709	1	17	0.109
Variance	ACI(0.05)	4.217	30	> 75	0.161
Variance	BR-CP	3.431	6	> 75	0.138

In the exploratory variance pulse, BR-CP has lower CD100 and earlier rolling-average first crossing than ACI(0.05), but both strict25 recovery estimates are censored in the observed 100-step window. Under a level pulse, ACI(0.05) has the lower CD100, and both rolling-average TTR values equal one. These rankings depend on the estimand: with $\varepsilon = 0$, the CD100 values are respectively 4.581 versus 6.267 for variance and 2.660 versus 2.210 for level (BR-CP versus ACI). A paired percentile bootstrap that resamples the *six datasets* as blocks (retaining their channels, repeated interventions, and method pairing) gives exploratory 95% intervals for CD100 differences (BR-CP minus ACI) of $[-1.134, -0.518]$ under variance and $[0.108, 0.744]$ under level at $\varepsilon = 0.03344$. Six dependent domain blocks are too few for strong inferential guarantees, and these intervals describe this post-hoc estimator rather than a registered confirmatory effect.

On variance-pulse channels, the *within-channel* BR-CP/ACI Winkler score ratio is below one for only 8 of 36 channels, with a median ratio 1.0146 (higher means worse for BR-CP). Raw mean Winkler scores are not scale-comparable across the illness, exchange and ETT datasets. Thus the supplementary experiment does *not* establish across-the-board efficiency superiority, and neither the simplified PI baseline nor the evaluated EWA-ACI variant should be misrepresented as exact reference implementations.

The full original GPU result, corrected portable CPU replay, per-event error traces, hash manifest, numerical tests, detector grid, epsilon sensitivity, and dataset-block bootstrap are archived as *separately versioned* exploratory evidence. A newly chosen hold-out, independent training-corpus audit, and full published baseline fidelity remain outstanding. No results from this exploratory checkpoint arm are mixed with the historical cross-domain forecaster's main tables.

11 Independent-seed detector null diagnostic and source fidelity

Pre-frozen diagnostic design (separate from Chronos). To probe whether our detector-space comparison remains fair on genuinely new data, we created a separate *detector-only* synthetic audit with independent seeds, a correctly specified stationary Gaussian AR(1) null ($\phi = 0.6$), and persistent level and innovation-variance changes. Each series has 1,100 observations, with a 300-step detector warm-up, 250-point initial anchor, 60-point post-alarm re-anchoring, two-sided standardized/winsorized Page increments ($k = 0.8$), and 800 monitored steps. The level intervention adds $(1 - \phi)\, 2$ to innovations starting at $t = 650$, while the variance intervention triples innovation scale; when these interventions are absent, the data-generating law is exactly stationary. Residual scores use the frozen, correctly specified AR(1) one-step predictor. A fixed grid was selected using 90 independent *development* seeds to target 0.5 alarms per 800 null steps. The selected thresholds ($h_{\mathrm{score}} = 5$, $h_{\mathrm{data}} = 16$) were then evaluated *once* on 300 disjoint test seeds, without retuning. This is not a Chronos or BR-CP interval-coverage experiment.

Table 11: New, independent-seed AR(1) detector-only diagnostic. The prespecified alarm-rate difference tolerance is 0.20 alarms per 800 steps; observed test difference 0.3533 *fails* that matching gate. Post-shift hit rate is at least one alarm in the first 120 steps. These hit rates are descriptive, *not* a false-alarm-matched power comparison.

Detector	Null alarms / 800	Level hit rate	Variance hit rate
Score-space ($h = 5$)	0.5300	0.7067	0.9800
Data-space ($h = 16$)	0.1767	0.9833	0.9733

The test null alarm-rate gap (score minus data) was 0.3533 per 800 steps; a paired seed bootstrap (5,000 draws) gave an exploratory 95% percentile interval $[0.250, 0.453]$. Thus even with independent development selection, a true no-change process and fresh test seeds, the empirical matching target did *not* transport to the hold-out. We did not reselect thresholds using test results. Accordingly, the post-change hit-rate comparison cannot isolate monitoring-space effects at equal false-positive operation. The variance-pulse conditional median delay among detected streams was 3 steps for score-space and 24 for data-space, whereas under a level shift score-space had the lower observed hit probability; these are ancillary descriptive quantities under *unequal* test alarm budgets. This test does not prove any universally superior detector. The frozen protocol, raw per-seed outcomes, explicit code and a 150-case equivalence test against the original Python detector are included with the supplementary audit.

Sequential calibration-stability follow-up (exploratory). The previous 90-development-seed experiment failed to transfer its false-alarm budget to the first test sample. We therefore ran a *sequential exploratory follow-up*, specifying its new seeds and threshold grid before inspecting each new outcome but *after* observing the first failure. A larger calibration sample (1,200 new development seeds) selected $h_{\text{score}} = 5$ and $h_{\text{data}} = 13$; on 900 disjoint new stationary-AR(1) null seeds these yielded 0.5344 and 0.4822 alarms per 800 steps, a difference of 0.0522 (paired-bootstrap 95% interval $[-0.018, 0.122]$), within that follow-up's stated tolerance of 0.15. Without any further threshold adjustment, a third independent set of 1,000 null seeds yielded 0.524 and 0.492 alarms per 800 steps (gap 0.032; bootstrap difference interval $[-0.027, 0.093]$). Thus larger *development* samples improved calibration stability for this particular exactly stationary synthetic null. These follow-up choices were motivated by inspecting the earlier result and are *not* an a priori registration of the entire sequence of studies.

On the third set of independently seeded persistent shifts, the 120-step detection proportions were 0.984 (score) versus 0.977 (data) for innovation-variance changes, with a paired hit-rate difference of 0.007 (bootstrap interval $[-0.005, 0.019]$); the conditional median detection delays were 3 versus 17 steps. For persistent level changes the hit rates were 0.687 versus 0.974, with a difference of -0.287 (bootstrap interval $[-0.316, -0.258]$) and conditional median delays of 44 versus 17 steps. Delay medians are conditional on an alarm, so they should not be interpreted independently of detection probability. These results qualify the central scientific intuition: monitoring conformity scores may respond rapidly to variance-driven competence shifts, but can be less sensitive to an observable level change. It is not universally superior to monitoring observations. This sequential series is *detector-only* and does not establish matched operating characteristics on the real or Chronos score streams. The original failure, larger-calibration results, frozen follow-up thresholds, raw new-seed results and both bootstrap files are retained, including the unfavorable level-shift outcomes.

Reference-algorithm and checkpoint provenance audit. The implementation tagged `AgACI` in the historical archive performs interval-score exponential-weight aggregation of ACI experts. The published AgACI of Zaffran et al. [10] instead aggregates interval boundaries using an online-expert algorithm such as Bernstein Online Aggregation and a quantile-oriented loss. Hence these methods must not be equated. Likewise, the historical `ConformalPID` code is a hand-specified quantile tracker with a bounded tanh integrator, omitting the learned scorecaster of Angelopoulos et al. [5], and is better called a *simplified PI-like comparator* rather than a source-equivalent full PID replication. These archived names are preserved solely for replay compatibility; numerical claims apply only to the implemented variants. The official Chronos-Bolt Small model card (https://huggingface.co/amazon/chronos-bolt-small) refers to zero-shot evaluation on 27 benchmark datasets. The original Chronos paper (Ansari et al., 2024, Table 3, https://cdn.amazon.science/1c/b5/1ec660274e52ac843a988c6065dc/chronos-learning-the-language-of-time-series.pdf) explicitly

places ETT (hourly/15-minute) and Exchange Rate in its zero-shot-evaluation group, supporting the stated study-level separation of those five benchmark names. This is not a byte-level audit of the exact THUML CSVs or the released Chronos-Bolt Small checkpoint's training shards; national illness is not among the listed 27 families. Consequently, strict six-dataset pretraining disjointness remains unverified.

11.1 Exploratory strongly adaptive and scorecasting comparators

As an additional response to the requests for strongly adaptive and scorecasting baselines, we replayed the *existing, unchanged* Chronos-Bolt Small forecast caches with two new calibration methods. The first is a one-step, scores-only port of the core SAOCP learner (online active experts, scale-free gradient steps, and coin-betting aggregation) from the public implementation of Bhatnagar et al. [11]. Its wrapper was adapted to accept the one-step absolute-error stream directly rather than the authors' original forecaster interface; it is *not* a byte-identical run of the published library. We initialized on the 300 warm-up scores, used $\alpha = 0.1$, expert lifetime factor 8, and prequentially updated only after each score was observed. The score-stream port and source attribution are included in supplementary code.

The second method uses the proportional update and log-tangent integrator described by Angelopoulos et al. [5], with an independently learned one-step AR(1) *scorecaster* fit on a rolling 300-point window of already observed scores. This is an explicitly labeled *PID–AR(1) variant*, not the authors' published Theta scorecaster configuration. All six development streams available in the frozen cache were unperturbed, distinct evaluation channels (`col000`); no future scores or evaluation channels entered hyperparameter selection. A fixed development grid over $\eta \in \{0.01, 0.03, 0.1\}$ and $K_I \in \{0, 0.1, 0.25\}$ selected $(\eta, K_I) = (0.1, 0)$, meaning the unrestricted scorecaster model *omitted* integral control. A separately reported, deliberately constrained $K_I > 0$ follow-up chose $(0.1, 0.1)$ on the same development streams; this post-hoc restriction is not a new prospective validation. Both use $C_{\text{sat}} = 40$ and project negative interval radii to zero; that projection precludes transfer of an unmodified theoretical PID coverage claim.

Table 12: Post-hoc Chronos score-only strongly adaptive and causal PID–AR(1) comparisons on 36 evaluation channels for each scenario. Winkler ratios are calculated *within each channel* against the corrected BR-CP and averaged over channels; values below one mean lower observed interval score, not proven statistical equivalence. SAOCP is a source-derived port; PID–AR(1) is a declared variant.

Scenario	Comparator	Coverage	Winkler/BR-CP	BR-CP coverage
No injected break	SAOCP score port	0.9208	0.9836	0.9001
No injected break	PID–AR(1), $K_I = 0.1$	0.8964	0.9575	0.9001
Level pulse	SAOCP score port	0.9204	0.9879	0.9007
Level pulse	PID–AR(1), $K_I = 0.1$	0.8963	0.9627	0.9007
Variance pulse	SAOCP score port	0.9155	0.9572	0.8997
Variance pulse	PID–AR(1), $K_I = 0.1$	0.8927	1.0505	0.8997

On the injected-event windows, pooling the same 564 event windows but taking only the first 100 post-event steps, SAOCP's observed miscoverage was 0.0833 versus BR-CP's 0.1084 for level pulses, and 0.1606 versus 0.1394 for variance pulses. This does not estimate conditional coverage; it describes paired finite windows. The cluster-bootstrap 95% interval for the *dataset-macro* Winkler ratio (SAOCP/BR-CP) under variance pulses was $[0.9333, 0.9753]$, based on only six dataset blocks. This and the other cluster intervals are exploratory. Overall average coverage alone would obscure SAOCP's less favorable early variance-pulse miscoverage despite its generally smaller observed interval score. Conversely, SAOCP is more favorable under the level pulse. Therefore these newly evaluated comparators *do not* support a claim of uniform BR-CP superiority. The source-equivalent, published BOA/pinball AgACI and the authors' exact Theta/PID implementation are still not reproduced. The resulting limited comparison is not presented as a full published-algorithm benchmark.

12 Limitations

Published-checkpoint scope. The original main synthetic and real-data arms use frozen linear forecasters and do not demonstrate performance on published TSFMs. Section 10 provides a separately labeled, exploratory Chronos-Bolt Small experiment, corrected post-hoc from cached forecasts. It does not establish

training/evaluation data disjointness, full published baseline equivalence, or a fresh confirmatory test on a novel hold-out; the extra SAOCP/PID–AR(1) source-derived comparisons are exploratory and not byte-identical runs of published reference pipelines.

Injected rather than dated breaks. Coverage debt needs a known τ and many replications, which injection supplies at the cost of assuming the break family. The event-local replay shows delayed alarm responses in five of eight Lehman channels, but none in four H1N1 channels within 60 weeks. Neither event has a same-observation alarm. The calendar dates are not proven statistical break times and the local observed hit rates are not conditional-coverage estimates. Finding real, dated events that *do* produce a conformity-score jump, most likely at longer horizons or in series whose conditional scale is not absorbed by instance normalization, is an important remaining test of the method.

Marginal, not conditional. BR-CP is evaluated for marginal coverage and does not establish conditional validity. Under the seasonal-amplitude break strict TTR-C is censored for the evaluated variants, including the known-break-time benchmark: violations cluster at seasonal peaks under these unconditioned-radius rules. Normalized or Mondrian scores conditioned on a regime variable are the natural remedy and compose with BR-CP.

No unconditional long-run guarantee is established. Because BR-CP uses clipping, time-varying steps, resets, and detector re-anchoring, the standard fixed-step ACI guarantee does not transfer directly. Empirically, the 21-cell mean marginal coverage is 0.8866 and the worst observed trend setting falls to 0.8647; the gap to the nominal 0.900 level is material and heterogeneous.

Closely spaced breaks. Section 8 shows BR-CP losing to ACI(0.05) on coverage debt when a second break lands inside the 60-step re-anchoring window. We do not study gradual drift regimes or adversarial sequences designed to exploit this. The trend family is the only one with a persistent change, and there BR-CP raises 8.4 alarms per series, which is correct behavior but a regime in which the reset mechanism is doing continuous work rather than responding to an event.

One-step horizon. Multi-horizon calibration requires either per-step layers or a joint construction; the interaction with detection is untested.

Incomplete implementation and external-event checks. The new synthetic matched-development detector study did not achieve identical held-out no-break false-alarm rates. The ACI boundary check shows that using original unbounded quantiles changes coverage by producing many infinite intervals. BR-CP projection and Wilks fallback frequencies were subsequently audited across all 21 original synthetic cells at $h = 6.0$, and the recovered 35-series real variance arm has now been independently instrumented for BR-CP and ACI(0.05). The recovered exchange-rate and illness files allow a new local event trace replay, including a correction to the archived alarm-delay origin, but the provided electricity file has the wrong benchmark schema and full pretraining-from-scratch remains unverified. The event traces are observed one-path trajectories rather than conditional-coverage estimates. Finally, published BOA/pinball AgACI and the authors' exact Theta/PID package were not rerun; a separately disclosed source-core SAOCP score port and PID–AR(1) scorecaster variant were replayed on the saved Chronos caches. All claims concern the explicitly disclosed implementations.

13 Conclusion

Marginal coverage alone can conceal transient undercoverage after a structural break. This study evaluated break-conditional coverage debt and TTR-C, provided a conditional finite-horizon adaptation-lag calculation for fixed-step ACI, and evaluated a score-triggered calibration wrapper that changes calibration behavior without retraining the forecaster. Across the evaluated simulations, no fixed step in the tested grid matched BR-CP's mean coverage debt, although the advantage was largely concentrated in trend breaks and fixed-step selection was retrospective. In controlled stress tests on five real-world benchmark datasets, the advantage over a fast, ex post tuned ACI baseline was smaller than in simulation: BR-CP recovered coverage more quickly with a small observed interval-score difference that was not statistically significant in the reported test. Replayed local diagnostics correct a historical alarm-offset error: seven delayed post-Lehman alarms occur across five of eight currency channels, whereas none occur in the four tested influenza channels during the 60-week post-H1N1 window. Neither event yields a contemporaneous alarm, and these observational traces do not establish break-conditional coverage or a causal detector effect.

The initial independent-seed synthetic detector null check failed its prespecified held-out alarm-rate matching tolerance; a sequential follow-up using substantially more development data matched observed null alarm frequencies on separate new AR(1) test seeds, but showed strong level-shift dependence and does not establish general monitoring-space superiority. The post-hoc published-checkpoint audit also showed that corrected BR-CP reduced early-window coverage debt under variance pulses but not under level pulses, and that its interval efficiency did not uniformly exceed tuned ACI. These exploratory observations do not upgrade the original evidence to a confirmatory foundation-model benchmark.

Newly added source-derived SAOCP and causal scorecaster comparisons further qualify the claim: SAOCP achieved higher overall coverage and lower average interval score than BR-CP in the evaluated variance-pulse Chronos arm, while BR-CP had lower miscoverage in the initial 100-step variance-pulse windows. These baseline add-ons were post-hoc and do not constitute source-identical evaluations of all requested original packages.

The results support reporting break-conditional diagnostics when post-break performance is operationally relevant. They also identify limitations that should be tested directly in future work, including independent new hold-outs and additional published foundation-model checkpoints, multi-horizon calibration, conditional rather than marginal coverage, and closely spaced breaks.

Reproducibility. Online Resource 1 contains the anonymous replication package, including code, per-replication outputs, result tables, figure sources, and scripts for the real-data pipeline. Public benchmark data are referenced by source rather than redistributed. The new synthetic audit scripts and raw test outputs are under `revision_analysis`; they are separate from and do not overwrite the historical main tables. The exchange-rate and influenza inputs were subsequently recovered and replayed using the frozen model; the archived 84-row case-study table was reproduced exactly, and 32,480 local trace records were exported. Input hashes, corrected alarm offsets, provenance warnings, and Figure 7 are included as distinct supplemental artifacts. These are descriptive realizations, not estimates of conditional coverage. The unrelated Electricity file encountered during Drive recovery was flagged and is not used to claim a new model-pretraining reproduction. The separately versioned Chronos-Bolt exploratory audit (Section 10) uses its own original GPU cache and corrected, CPU-replayed results; no erroneous pre-correction summary is used for the new analysis.

Declarations

Funding. No funding was received for conducting this study.

Competing interests. The authors have no relevant financial or non-financial interests to disclose.

Ethical approval

Not applicable. This computational study used simulations and publicly available benchmark time-series; no research involving human participants or animals was performed.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Clinical Trial Number. Not applicable.

Data availability. The target benchmark datasets are publicly available from the THUML Time-Series-Library; recovered target-file SHA-256 values and derived analyses are documented in Online Resource 1. The full source-equivalent reconstruction of the archived global forecaster's pretraining corpus was not completed: a recovered electricity file had a different benchmark schema and must not be represented as the original Electricity benchmark. This limitation is reported in the scientific audit.

Code availability. The code used for simulation, calibration, analysis, and figure generation is included in Online Resource 1.

Use of generative AI. ChatGPT (OpenAI) assisted with language editing, manuscript restructuring, and analysis-code review. The authors remain responsible for verifying citations, computations, interpretation, and the submitted manuscript.

References

[1] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. P. Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024.

[2] A. Das, W. Kong, R. Sen, and Y. Zhou. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 41st International Conference on Machine Learning*, PMLR 235:10148–10167, 2024.

[3] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo. Unified training of universal time series forecasting transformers. In *Proceedings of the 41st International Conference on Machine Learning*, PMLR 235:53140–53164, 2024.

[4] I. Gibbs and E. Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, volume 34, pages 1660–1672, 2021.

[5] A. N. Angelopoulos, E. J. Candès, and R. J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems*, volume 36, pages 23047–23074, 2023. https://doi.org/10.52202/075280-1000.

[6] H. Papadopoulos, K. Proedrou, V. Vovk, and A. Gammerman. Inductive confidence machines for regression. In *Machine Learning: ECML 2002*, pages 345–356. Springer, 2002. https://doi.org/10.1007/3-540-36755-1_29.

[7] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018. https://doi.org/10.1080/01621459.2017.1307116.

[8] C. Xu and Y. Xie. Sequential predictive conformal inference for time series. In *Proceedings of the 40th International Conference on Machine Learning*, PMLR 202:38707–38727, 2023.

[9] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics*, 51(2):816–845, 2023. https://doi.org/10.1214/23-AOS2276.

[10] M. Zaffran, O. Féron, Y. Goude, J. Josse, and A. Dieuleveut. Adaptive conformal predictions for time series. In *Proceedings of the 39th International Conference on Machine Learning*, PMLR 162:25834–25866, 2022.

[11] A. Bhatnagar, H. Wang, C. Xiong, and Y. Bai. Improved online conformal prediction via strongly adaptive online learning. In *Proceedings of the 40th International Conference on Machine Learning*, pages 2337–2363, 2023.

[12] I. Gibbs and E. J. Candès. Conformal inference for online prediction with arbitrary distribution shifts. *Journal of Machine Learning Research*, 25(162):1–36, 2024.

[13] S. Sun and R. Yu. Conformal prediction for time-series forecasting with change points. In *Advances in Neural Information Processing Systems*, volume 38, 2025.

[14] V. Vovk, I. Petej, I. Nouretdinov, E. Ahlberg, L. Carlsson, and A. Gammerman. Retrain or not retrain: Conformal test martingales for change-point detection. In *Proceedings of the Tenth Symposium on Conformal and Probabilistic Prediction and Applications*, PMLR 152:191–210, 2021.

[15] E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1–2):100–115, 1954. https://doi.org/10.1093/biomet/41.1-2.100.

[16] G. Lorden. Procedures for reacting to a change in distribution. *Annals of Mathematical Statistics*, 42(6):1897–1908, 1971. https://doi.org/10.1214/aoms/1177693055.

[17] S. S. Wilks. Determination of sample sizes for setting tolerance limits. *Annals of Mathematical Statistics*, 12(1):91–96, 1941. https://doi.org/10.1214/aoms/1177731788.